

Martingale sampling diagnostics

Exact-zero tests and practical-equivalence analysis

ARC-Challenge, ARC-Easy and CommonsenseQA | GPT-4o mini and DeepSeek V4 Pro

Prepared for an initial methodological review

17 September 2026

Purpose. This note updates the sampling diagnostics with an explicit tolerance. The default is $\epsilon = 0.001$: a class-wise mean residual within ± 0.001 is treated as practically negligible only when a formal equivalence test supports that conclusion. This is an initial methodological readout, not a final model-level claim.

Main observations

- The average local residual magnitude is very small in five of the six runs. The exception is GPT-4o mini on CommonsenseQA ($1.79e-03$), about 838 times the DeepSeek value on the same selected questions.
- Exact equality and practical equivalence give different answers: 23/30 class-wise bootstrap intervals exclude zero and 6/30 t-tests survive within-run Bonferroni correction, while 30/30 class means pass TOST equivalence at ± 0.001 .
- Using the one-sided 95% trajectory-bootstrap upper bound, 5/6 runs have mean L2 residual below $\epsilon_{L2} = 0.001$. This is the more meaningful aggregate assessment because signed residuals cancel across classes by construction.
- The model probabilities are strongly saturated. Mean maximum probabilities range from 0.998364 to 1.000000. The current experiment mainly examines near-one-hot states rather than the full probability simplex.

1. Experimental coverage

For each visited history, the experiment computes the conditional expected next probability vector and subtracts the current vector: $r_t = E[p_{t+1} | F_t] - p_t$. The saved tensors contain 49 transitions and five classes. Inference in the notebook first averages dependent steps within a trajectory and then resamples complete trajectories.

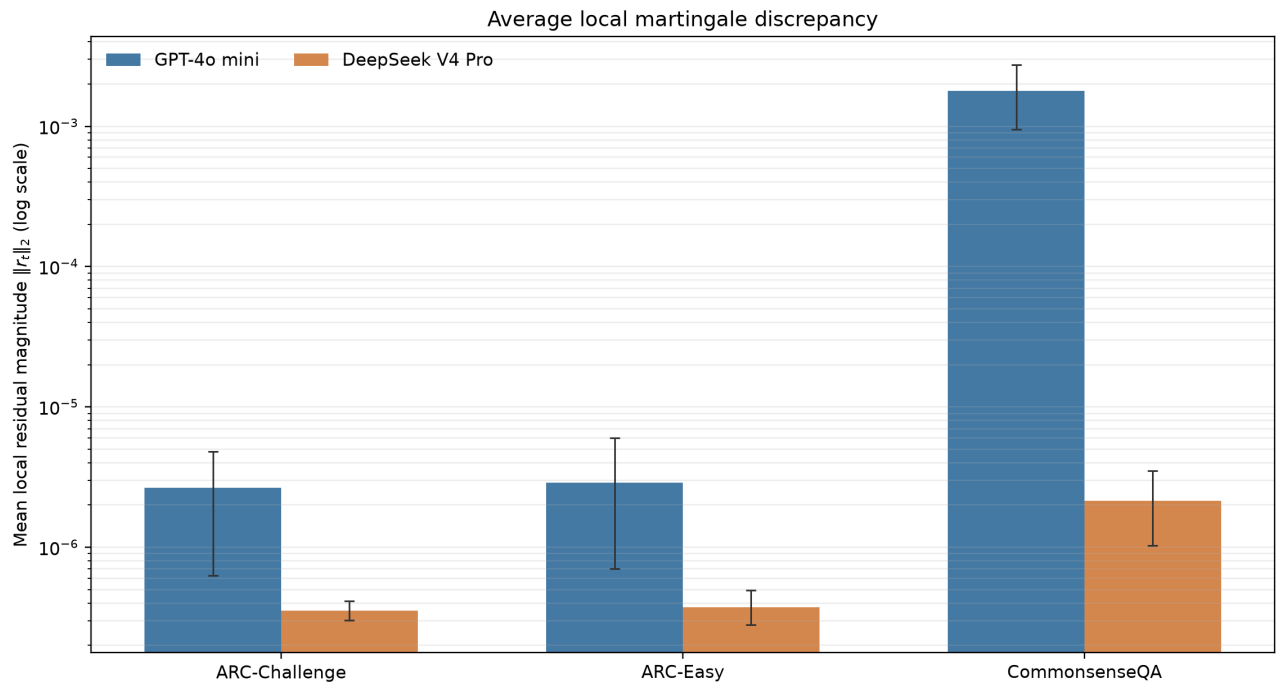
Dataset	Model	Questions	Traj./question	Clusters	States	Incorrect
ARC-Challenge	GPT-4o mini	5	5	25	1225	0
ARC-Challenge	DeepSeek V4 Pro	5	5	25	1225	0
ARC-Easy	GPT-4o mini	5	5	25	1225	0
ARC-Easy	DeepSeek V4 Pro	10	5	50	2450	0
CommonsenseQA	GPT-4o mini	5	5	25	1225	0
CommonsenseQA	DeepSeek V4 Pro	5	5	25	1225	0

The selected question indices match across models for ARC-Challenge (True) and CommonsenseQA (True), but not for ARC-Easy (False). ARC-Easy also uses 10 questions for DeepSeek and 5 for GPT-4o mini, so that model comparison is descriptive rather than paired.

Tests implemented in the notebook

- Class-wise mean signed residual with a 95% percentile bootstrap interval, resampling complete trajectories.
- Two-sided one-sample t-test of trajectory-mean residuals against zero, with a Bonferroni threshold across five classes.
- Class-wise TOST equivalence test for the interval $(-0.001, +0.001)$; at $\alpha = 0.05$ this corresponds to a 90% equivalence confidence interval.
- An additional signed-residual t-test after averaging over classes, reported as requested but interpreted only as a probability-conservation check.
- A practical aggregate criterion for the non-negative L2 residual: its one-sided 95% bootstrap upper bound must be below $\epsilon_{L2} = 0.001$.
- L2 residual magnitude by step, including mean, median and pointwise trajectory-bootstrap intervals.
- Expected-next versus current probability, matching the active visualization in the notebook.

2. Overall residual magnitude

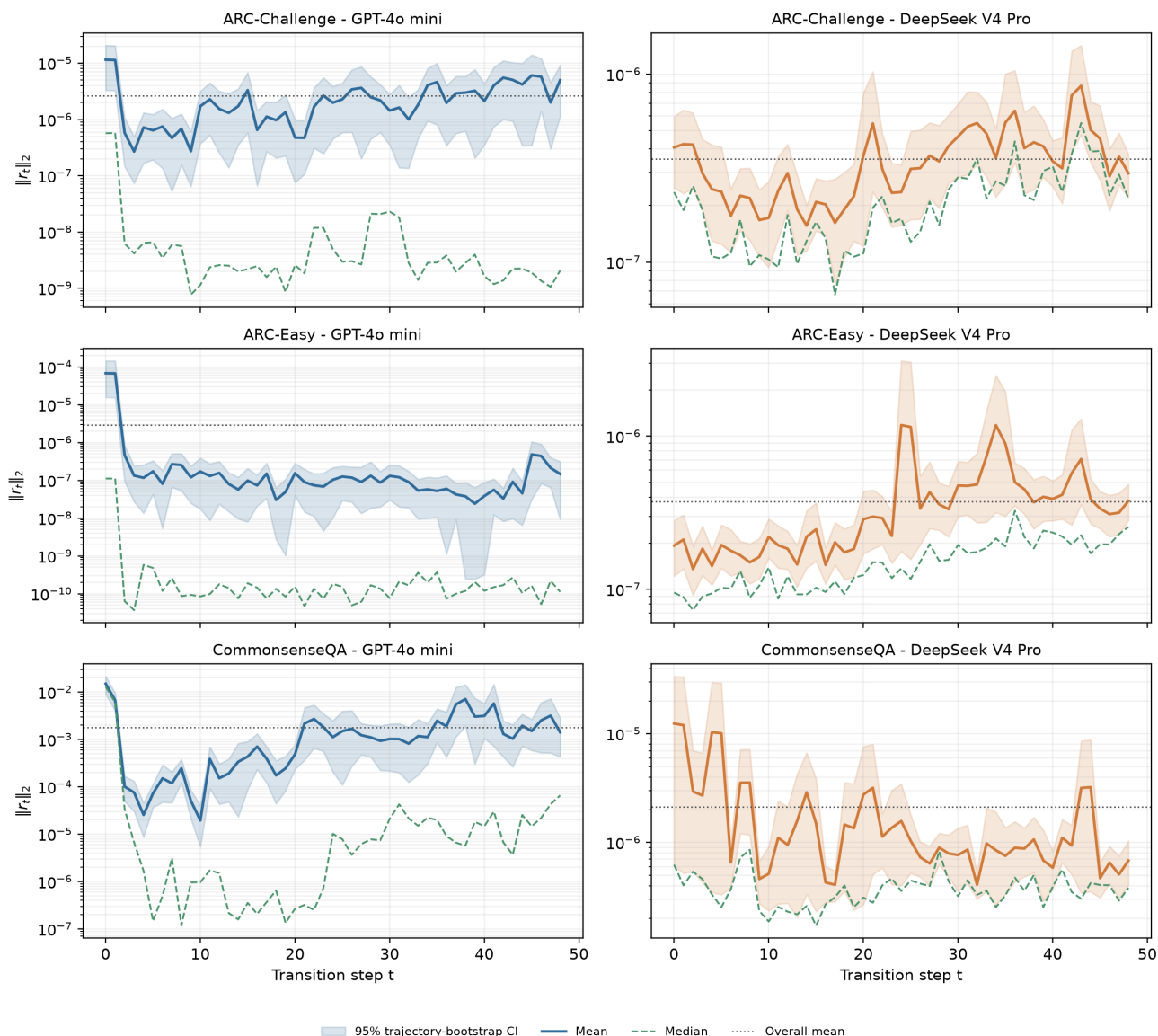


Dataset	Model	Mean L2	95% CI	Median	Maximum
ARC-Challenge	GPT-4o mini	2.656e-06	[6.31e-07, 4.80e-06]	2.91e-09	8.32e-05
ARC-Challenge	DeepSeek V4 Pro	3.539e-07	[3.02e-07, 4.12e-07]	1.86e-07	6.30e-06
ARC-Easy	GPT-4o mini	2.885e-06	[7.00e-07, 6.03e-06]	1.46e-10	8.03e-04
ARC-Easy	DeepSeek V4 Pro	3.737e-07	[2.81e-07, 4.93e-07]	1.38e-07	4.67e-05
CommonsenseQA	GPT-4o mini	1.789e-03	[9.49e-04, 2.75e-03]	6.08e-06	1.02e-01
CommonsenseQA	DeepSeek V4 Pro	2.135e-06	[1.03e-06, 3.51e-06]	3.49e-07	2.57e-04

The median is well below the mean in every run, especially for GPT-4o mini. This indicates a right-skewed distribution: most transitions have negligible discrepancy, with a small number of larger excursions driving the average. CommonsenseQA with GPT-4o mini stands apart both in its mean and its maximum residual.

3. Evolution over the trajectory

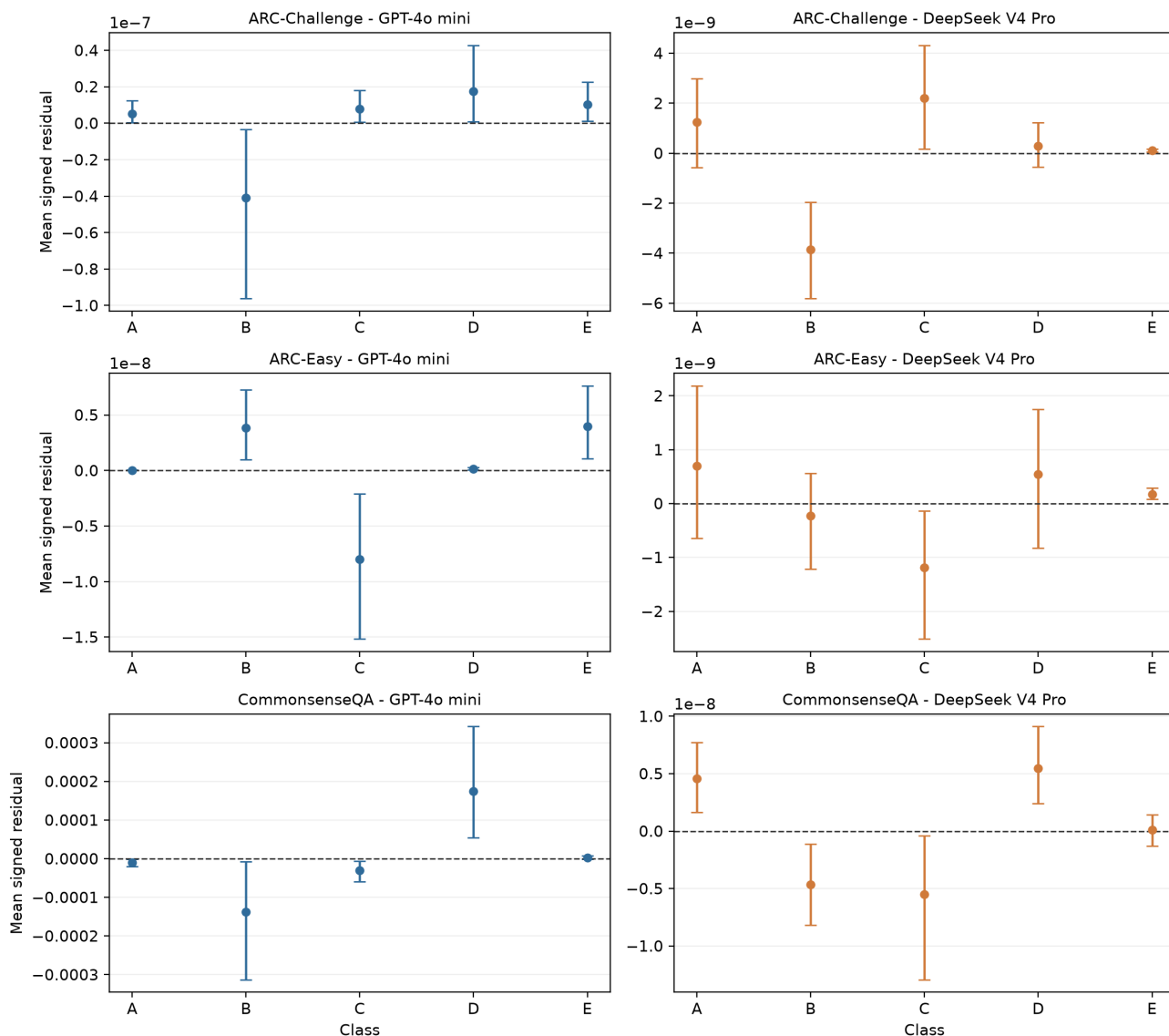
Residual magnitude over the 49 transitions



The log scale is intentional: the runs differ by several orders of magnitude. There is no common monotone increase across all six panels. Instead, the means are driven by intermittent step-specific spikes. The median often remains close to the numerical floor while the mean moves, which again points to a small subset of trajectories or histories accounting for most of the observed discrepancy.

4. Are mean signed residuals different from zero?

Class-wise mean signed residuals with 95% trajectory-bootstrap intervals



Across 30 dataset-model-class combinations, 23 percentile bootstrap intervals exclude zero and 6 t-tests remain significant after within-run Bonferroni correction. This answers the literal question of exact equality. It does not answer whether the departures are large enough to matter.

At epsilon = 0.001, 30/30 class means are statistically equivalent to zero by TOST. A tiny confidence interval such as $[4.27e-10, 1.23e-08]$ can therefore exclude exact zero and still sit comfortably inside the tolerance band. These statements are not contradictory: the first concerns exact equality; the second concerns practical size.

5. Aggregate tests and epsilon sensitivity

Dataset	Model	Signed mean	Exact p	TOST p	Mean L2	L2 upper 95%	< .001?
ARC-Challenge	GPT-4o mini	2.44e-18	9.53e-04	0.00e+00	2.66e-06	4.40e-06	yes
ARC-Challenge	DeepSeek V4 Pro	1.14e-18	3.33e-02	0.00e+00	3.54e-07	4.01e-07	yes
ARC-Easy	GPT-4o mini	2.48e-18	7.65e-05	0.00e+00	2.89e-06	5.45e-06	yes
ARC-Easy	DeepSeek V4 Pro	1.49e-18	3.75e-04	0.00e+00	3.74e-07	4.70e-07	yes
CommonsenseQA	GPT-4o mini	5.44e-19	3.31e-01	0.00e+00	1.79e-03	2.59e-03	no
CommonsenseQA	DeepSeek V4 Pro	1.05e-18	8.41e-02	0.00e+00	2.13e-06	3.26e-06	yes

How to read TOST and the tolerance decision

TOST means *Two One-Sided Tests*. Instead of asking whether the mean is exactly zero, it asks whether the mean is convincingly inside a pre-specified negligible interval, here $(-0.001, +0.001)$. One test checks that the mean is above the lower boundary and the other checks that it is below the upper boundary. Equivalence is concluded only when both tests reject their boundary null hypothesis at $\alpha = 0.05$. Equivalently, the entire 90% confidence interval must lie inside the tolerance interval.

The **< .001?** column concerns the non-negative L2 residual, not the signed mean. A 'yes' means the one-sided 95% trajectory-bootstrap upper confidence bound for the mean L2 residual is below 0.001. A 'no' means the data do not establish that the mean L2 error is smaller than this tolerance. It does not mean that the residual is larger than 0.001 at every transition.

The class-aggregated signed test is almost guaranteed to return zero: both the current and expected-next vectors sum to one, so their component-wise residuals sum to zero. It is included because it was requested, but it should not be treated as evidence for the martingale property. The L2 column avoids this cancellation and is the useful aggregate result.

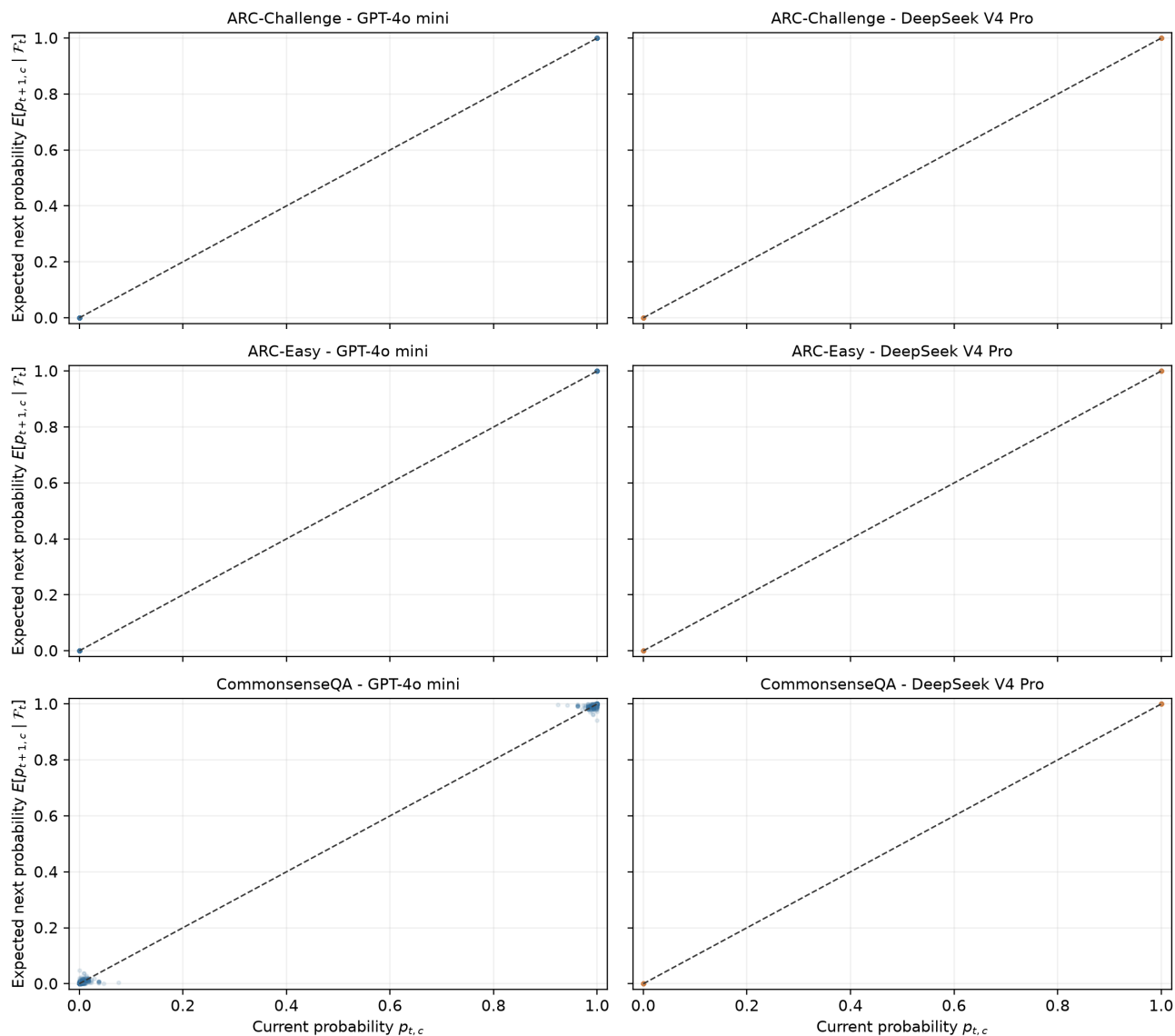
Sensitivity of the aggregate L2 conclusion

Dataset	Model	epsilon=0.0001	epsilon=0.001	epsilon=0.01
ARC-Challenge	GPT-4o mini	pass	pass	pass
ARC-Challenge	DeepSeek V4 Pro	pass	pass	pass
ARC-Easy	GPT-4o mini	pass	pass	pass
ARC-Easy	DeepSeek V4 Pro	pass	pass	pass
CommonsenseQA	GPT-4o mini	fail	fail	pass
CommonsenseQA	DeepSeek V4 Pro	pass	pass	pass

A run passes when the one-sided 95% bootstrap upper bound for its trajectory-mean L2 residual is below epsilon. The table makes the scientific consequence of the tolerance choice explicit; epsilon should ultimately be justified by a downstream probability or decision effect, rather than selected after seeing significance.

6. Expected-next versus current probability

Expected next probability versus current probability



Points on the dashed diagonal have zero class residual. Most observations lie at the corners because the saved probabilities are nearly one-hot. Departures are visually most noticeable for GPT-4o mini on CommonsenseQA, consistent with the L2 summary.

7. Dataset-level reading

ARC-Challenge

On matched question indices, GPT-4o mini has a mean L2 residual of $2.66\text{e-}06$ and DeepSeek $3.54\text{e-}07$ (ratio 7.5). Both remain very small in absolute terms and all evaluated states are correct.

ARC-Easy

Both models show very small discrepancies ($2.89\text{e-}06$ for GPT-4o mini and $3.74\text{e-}07$ for DeepSeek). However, the questions are not matched and DeepSeek uses twice as many questions, so the ratio should not be read as a controlled model comparison.

CommonsenseQA

GPT-4o mini has a mean L2 residual of $1.79\text{e-}03$, compared with $2.13\text{e-}06$ for DeepSeek (ratio 838). This is the clearest separation in the current data and occurs on matched question indices. GPT-4o mini is also less saturated here (minimum maximum probability 0.924), giving the residual more room to vary.

Provisional conclusion

At the pre-specified working tolerance of 0.001 per signed class residual, the class-wise results are practically close to zero even though several are distinguishable from exact zero. The aggregate L2 criterion is stricter and identifies any runs whose overall discrepancy is not below 0.001. These findings are compatible with approximate martingale behavior only in the sampled, highly confident states; they do not establish the full conditional martingale property. GPT-4o mini on CommonsenseQA remains the clearest case for follow-up.

Recommended next run

- Increase the number of questions substantially and use identical question indices for both models on every dataset.
- For population-level uncertainty, resample questions as the outer cluster and trajectories within questions, rather than treating all question-trajectory pairs as fully independent.
- Pre-specify epsilon using a downstream consequence, such as the smallest probability shift that can change calibration or a decision, and retain the sensitivity table.
- Inspect the largest CommonsenseQA GPT-4o mini residuals at the prompt/history level to determine whether they arise from a reproducible behavioral transition or from provider/log-probability truncation.

Appendix A. Full class-wise test results

Dataset	Model	Class	Mean	Bootstrap 95% CI	Exact p	Bonf.	TOST p	Equiv.
ARC-Challenge	GPT-4o mini	A	5.273e-09	[4.27e-10, 1.23e-08]	1.13e-01	no	3.96e-117	yes
ARC-Challenge	GPT-4o mini	B	-4.086e-08	[-9.61e-08, -3.41e-09]	1.11e-01	no	7.89e-96	yes
ARC-Challenge	GPT-4o mini	C	7.734e-09	[7.30e-10, 1.80e-08]	1.12e-01	no	3.79e-113	yes
ARC-Challenge	GPT-4o mini	D	1.752e-08	[9.46e-10, 4.28e-08]	1.32e-01	no	4.76e-104	yes
ARC-Challenge	GPT-4o mini	E	1.033e-08	[1.06e-09, 2.26e-08]	8.17e-02	no	3.86e-111	yes
ARC-Challenge	DeepSeek V4 Pro	A	1.254e-09	[-5.73e-10, 2.98e-09]	1.88e-01	no	4.49e-130	yes
ARC-Challenge	DeepSeek V4 Pro	B	-3.857e-09	[-5.81e-09, -1.96e-09]	8.04e-04	yes	3.45e-129	yes
ARC-Challenge	DeepSeek V4 Pro	C	2.201e-09	[1.79e-10, 4.32e-09]	5.38e-02	no	2.10e-128	yes
ARC-Challenge	DeepSeek V4 Pro	D	2.909e-10	[-5.59e-10, 1.23e-09]	5.42e-01	no	4.08e-137	yes
ARC-Challenge	DeepSeek V4 Pro	E	1.117e-10	[5.39e-11, 1.74e-10]	1.63e-03	yes	2.63e-165	yes
ARC-Easy	GPT-4o mini	A	2.035e-11	[8.84e-12, 3.43e-11]	5.62e-03	yes	1.91e-181	yes
ARC-Easy	GPT-4o mini	B	3.852e-09	[9.69e-10, 7.27e-09]	2.73e-02	no	4.14e-124	yes
ARC-Easy	GPT-4o mini	C	-7.988e-09	[-1.52e-08, -2.10e-09]	2.80e-02	no	1.88e-116	yes
ARC-Easy	GPT-4o mini	D	1.543e-10	[3.94e-11, 2.84e-10]	2.27e-02	no	5.16e-158	yes
ARC-Easy	GPT-4o mini	E	3.962e-09	[1.05e-09, 7.61e-09]	3.13e-02	no	1.57e-123	yes
ARC-Easy	DeepSeek V4 Pro	A	6.976e-10	[-6.48e-10, 2.18e-09]	3.41e-01	no	2.16e-261	yes
ARC-Easy	DeepSeek V4 Pro	B	-2.285e-10	[-1.21e-09, 5.64e-10]	6.22e-01	no	4.74e-271	yes
ARC-Easy	DeepSeek V4 Pro	C	-1.188e-09	[-2.51e-09, -1.40e-10]	5.93e-02	no	6.71e-265	yes
ARC-Easy	DeepSeek V4 Pro	D	5.455e-10	[-8.21e-10, 1.75e-09]	4.16e-01	no	3.09e-263	yes
ARC-Easy	DeepSeek V4 Pro	E	1.733e-10	[8.12e-11, 2.86e-10]	1.75e-03	yes	0.00e+00	yes
CommonsenseQA	GPT-4o mini	A	-9.525e-06	[-2.02e-05, -1.33e-06]	6.89e-02	no	2.22e-40	yes
CommonsenseQA	GPT-4o mini	B	-1.372e-04	[-3.13e-04, -8.14e-06]	1.03e-01	no	6.77e-11	yes
CommonsenseQA	GPT-4o mini	C	-3.070e-05	[-5.90e-05, -6.63e-06]	3.12e-02	no	6.80e-30	yes
CommonsenseQA	GPT-4o mini	D	1.747e-04	[5.43e-05, 3.42e-04]	3.40e-02	no	7.47e-11	yes
CommonsenseQA	GPT-4o mini	E	2.754e-06	[-1.55e-06, 7.43e-06]	2.55e-01	no	2.85e-48	yes
CommonsenseQA	DeepSeek V4 Pro	A	4.557e-09	[1.62e-09, 7.74e-09]	9.30e-03	yes	2.75e-124	yes
CommonsenseQA	DeepSeek V4 Pro	B	-4.633e-09	[-8.17e-09, -1.14e-09]	1.88e-02	no	6.56e-123	yes
CommonsenseQA	DeepSeek V4 Pro	C	-5.498e-09	[-1.29e-08, -4.10e-10]	1.12e-01	no	1.07e-116	yes
CommonsenseQA	DeepSeek V4 Pro	D	5.480e-09	[2.39e-09, 9.10e-09]	4.79e-03	yes	2.38e-123	yes
CommonsenseQA	DeepSeek V4 Pro	E	9.395e-11	[-1.28e-09, 1.43e-09]	8.93e-01	no	4.27e-133	yes

The p-values are two-sided. Because each returned p-value already combines both tails, it is compared with $\alpha = 0.05$, not $\alpha/2$. The $\alpha/2$ split appears in the endpoints of the corresponding two-sided confidence interval.

Bonf. reports the Bonferroni-corrected exact-zero decision. Five class-wise tests are performed within each dataset-model run. If every test used 0.05 independently, the chance of at least one false positive would exceed 5%. Bonferroni controls this family-wise error by comparing each exact p-value with $0.05/5 = 0.01$. 'yes' means the exact-zero null is rejected even under this more conservative threshold; 'no' means it is not.

Equiv. reports the practical-equivalence decision from TOST. 'yes' means both one-sided tests passed and the 90% confidence interval for the class mean lies entirely inside $(-0.001, +0.001)$. The residual can therefore be statistically different from exact zero while still being negligible at the chosen tolerance. 'no' means equivalence was not established; it does not by itself prove a meaningful violation.

Scope and reproducibility

Source files: the six *martingale_sampling_results.npz* artifacts under *data/results_mp_sampling*. Analysis follows the current sampling notebook with 20,000 bootstrap draws for class-wise and overall summaries and 10,000 draws for step plots. Random seed: 42. Default tolerances: `epsilon_class = epsilon_L2 = 0.001`. This document was generated directly from the saved arrays; no model calls were repeated.